

Machine Learning 89-511, Spring 2025

Assignment 1

Isaiah Kriegman

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ישעיה קריגמן

1 Correlation and Independence

- A. Let X and Y be two independent random variables. Prove that they are uncorrelated, i.e., $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, or $\text{Cov}(X, Y) = 0$.

Solution:

$$\begin{aligned}\mathbb{E}[XY] &= \int \int XY f_{XY}(X, Y) dX dY \\ &= \int \int XY f_X(X) f_Y(Y) dX dY \quad (\text{Independence}) \\ &= \int X f_X(X) dX \int Y f_Y(Y) dY \\ &= \mathbb{E}[X] \mathbb{E}[Y].\end{aligned}$$

- B. Give an example of two uncorrelated random variables that are not independent.

Solution:

Correlation only measure a linear relationship, so we can find a non-linear relationship to satisfy this. Take $X \sim \text{Unif}(-1, 1)$ and $Y = X^2$. Their linear relationships on either side of 0 “cancel out”:

$$\begin{aligned}\mathbb{E}[XY] &= \mathbb{E}[X^3] = 0 \\ \mathbb{E}[X]\mathbb{E}[Y] &= 0 \cdot \mathbb{E}[Y] = 0.\end{aligned}$$

- C. Give an example of three random variables such that every pair is independent, but any variable can be completely determined if the two others are known.

Solution:

X , Y are each independent fair coin flips and Z is the xor function over them. It's obvious that knowing any two of these determines the third. Now we need to show that, without loss of generality, X and Z are independent:

$$P(Z = 1, X = 1) = P(Z = 1|X = 1) \cdot P(X = 1) = 0.5 \cdot 0.5 = P(Z = 1) \cdot P(X = 1).$$

- D. Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be a set of column vectors of dimension $d > n$. Let $C = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top$ be the empirical population correlation matrix. Prove that C is positive semi-definite, i.e., for any $\mathbf{z} \in \mathbb{R}^d$, $\mathbf{z}^\top C \mathbf{z} \geq 0$.

Solution:

If you squint hard enough you can see that $C = XX^\top$. So

$$\mathbf{z}^\top XX^\top \mathbf{z} = (X^\top \mathbf{z})^\top X^\top \mathbf{z} = \sum (\text{squares}) \geq 0.$$

2 Multivariate Gaussians

Let $X \sim \mathcal{N}(\mu, \Sigma)$ be a standard multivariate Gaussian of dimension d , with $\mu = \mathbf{0}$ and $\Sigma = I_d$.

- A. For $d = 1$, what is the probability that a sample is within one standard deviation from the mean (zero)?

Solution:

In R:

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pnorm(1) - pnorm(-1)
# 0.6826895
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- B. Prove that as $d \rightarrow \infty$, the probability that a sample is within a unit distance from the center decays to zero.

Solution:

I don't have a formal proof but I can show it intuitively:

Given that $\Sigma = I_d$, we know that the components are independent. We are within one unit of 0 when $\sum_{i=1}^d X_i^2 \leq 1$. Note that $Z = \sum_{i=1}^d X_i^2 \sim \text{Chisq}(d)$. As d increased the density of Z get pushed more and more positive. Intuitively it is very unlikely that a sum of very many squared standard normals would be small. The higher the value of d the lower that probability. With $d = 1,000,000$ there's almost 0 chance that $Z < 1$.

3 Linear Transformation of the Covariance Matrix

Let $X = [X_1, \dots, X_d]$ be a random vector with covariance matrix C . Let $\mathbf{w} \in \mathbb{R}^d$ and define $y = \mathbf{w}^\top X$. Show that $\text{Var}(y) = \mathbf{w}^\top C \mathbf{w}$.

Solution:

We have that

$$\begin{aligned} \text{Var}(y) &= \text{Var}\left(\sum \mathbf{w}_i X_i\right) = \sum_{i,j=1} \text{Cov}(\mathbf{w}_i X_i, \mathbf{w}_j X_j) = \sum_{i,j=1} \mathbf{w}_i \mathbf{w}_j \text{Cov}(X_i, X_j) \\ &= \mathbf{w}^\top C \mathbf{w}. \end{aligned}$$

4 Eigen Decomposition

Let A be an $n \times n$ square matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$.

- a. Prove that A and $A^\top$ have the same set of eigenvalues.

Solution:

A and $A^\top$ have the same characteristic polynomials:

$$\det(A - \lambda I) = \det((A - \lambda I)^\top) = \det(A^\top - \lambda I)$$

- b. Assume A is diagonalizable. Prove that $\det(A) = \prod_{i=1}^n \lambda_i$.

Solution:

5 Gradients

For the following functions $f(\mathbf{x}) : \mathbb{R}^d \rightarrow \mathbb{R}$, state the range where they are differentiable and compute their gradients with respect to $\mathbf{x}$. Here, $\mathbf{w} \in \mathbb{R}^d$ is a constant vector.

a.
$$f(\mathbf{w}, \mathbf{x}) = \begin{cases} 0 & \text{if } \mathbf{w}^\top \mathbf{x} \geq 1 \\ 1 - \mathbf{w}^\top \mathbf{x} & \text{otherwise} \end{cases}$$

Range of differentiability:

Gradient:

b. $f(\mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}$. Show that $\nabla f(\mathbf{x}) = f(\mathbf{x})(1 - f(\mathbf{x}))\mathbf{w}$.

Range of differentiability:

Gradient: